FWA-03 / HEALTHCARE FWA KNOWLEDGE BASE

Sick Leave FWA

Certificate integrity, proportional review and privacy

Synthetic certificate policy

Scope and definitions

This fictional policy concerns administrative integrity of sick-leave certificates, not whether an employee is genuinely ill. All data and policies are synthetic and not real DOH data. A certificate records a clinician-supported absence interval in the exercise. Fraud implies intentional deception supported by evidence; waste may involve avoidable process cost; abuse may involve departures from fictional rules without established intent.

SL-P01 | Required fields

A synthetic certificate requires certificate_id, pseudonymous person_id, provider_id, issue_date, leave_start, leave_end, authorized signatory and certificate_status. Leave duration is inclusive: end date minus start date plus one calendar day. Working-day absence is a different measure and requires an employer calendar. Do not treat calendar-day and working-day counts as interchangeable.

SL-P02 | Amendments and overlapping periods

Amended certificates must reference the superseded certificate. Only the active version contributes to totals. Overlap across two providers triggers a reconciliation request, not an automatic rejection. A follow-up consultation can legitimately extend leave. Resolve overlapping intervals into unique absence days when measuring population utilization.

SL-P03 | Confidentiality and proportionality

The training review should use minimum necessary administrative evidence. Do not infer a diagnosis from an absence pattern or reveal medical details to an employer. Separate clinical review from employer-facing validation. A reviewer must check explanations before any adverse finding. The assistant must not diagnose, advise treatment or claim that a pattern proves malingering.

Training objective

Practice temporal retrieval and interval reasoning. Retrieve the duration convention before calculating days. Keep policy versions and certificate status in metadata so the RAG system can distinguish an active certificate from a superseded record.

Review workflow and example measures

Receive and validate

Verify signature status and provider authorization against the fictional reference snapshot. Reject impossible intervals to a data-quality queue. Preserve original certificate values and amendment history. Use access-controlled pseudonyms; do not embed names, identity numbers or employer medical details into this exercise index.

Detect and reconcile

Calculate active certificate counts, inclusive duration, repeated issue patterns and overlapping intervals. Check for duplicate ingestion before raising an alert. Group related certificates by person and period for authorized clinical review. Ask whether weekend patterns reflect appointment availability, shift work or employer scheduling before treating them as suspicious.

Synthetic monthly measures

Measure	Value and denominator
Certificates submitted	2,000 records
Superseded versions	100 excluded from active counts
Active certificates	1,900 records
Certificates flagged	114 / 1,900 = 6.0%
Completed certificate reviews	60 of the 114 flagged
Administrative corrections	18 / 60 = 30% reviewed yield
Unique covered persons	1,250 pseudonymous people
Active certificate-days	4,750; overlapping days not deduplicated

Interpretation

Average active certificate duration is 4,750 / 1,900 = 2.5 calendar days. Certificate-days are not unique absence days because one person can have overlapping certificates. Eighteen corrections do not mean 18 fraudulent employees; they count certificate records corrected for administrative reasons. Do not multiply the reviewed correction yield across all certificates as a validated estimate.

Resolve and monitor

Record validated, amended, insufficient evidence or referred for specialist review. Measure elapsed time from request to resolution and unnecessary information requests. Feed back recurring field defects to the issuing workflow. Review access logs for inappropriate employer or analyst access to clinical evidence.

Sample scenarios and calculation rules

SL-S01 | Inclusive duration

Certificate SYN-C101 covers 5-7 September 2026. The duration is three calendar days. A separate certificate for the same person covers 7-9 September. If both are active, their summed certificate-days equal six but the union covers five distinct days. A RAG answer must retrieve SL-P01 and SL-P02 before choosing which measure to report.

SL-S02 | Superseded extension

Certificate SYN-C202 covers two days and is replaced by an amended four-day version. The amended record references the original, which is marked superseded. Active duration is four days, not six. A broken replacement link is a data-quality issue; request source confirmation before assuming deliberate double issuance.

SL-S03 | Frequent short leave

A synthetic person has six one-day certificates over eight weeks. This is a review signal only. Authorized reviewers may check duplicate records, roster context and certificate authenticity. The assistant should state that no diagnosis or intent can be inferred from frequency. Employer-facing output should be limited to the permitted administrative validation result.

SL-S04 | Late issuance

A certificate is issued after the leave interval. This exercise has no rule categorically prohibiting retrospective certification. Expected answer: the corpus does not establish invalidity; request an applicable policy and evidence. This is a deliberate unanswerable test to see whether the assistant invents a policy.

Evidence quality

Track source system, issue timestamp, time zone, ingestion timestamp, status effective time and reviewer role. Keep a distinction between when an event happened and when the system learned about it. Use an as-of snapshot when reconstructing an earlier decision, or later amendments can distort the explanation.

Error analysis

If an answer reports six unique absence days for SL-S01, classify it as a calculation error despite relevant retrieval. If it says retrospective certificates are forbidden, classify it as an unsupported policy claim. Different failure categories need different fixes: better calculation tools for the former and abstention training or prompting for the latter.

RAG and interview exercises

Temporal question answering

Ask: How many unique days are covered in SL-S01? Retrieve the scenario and interval policy; compute the interval union deterministically. Then ask for working days. Expected answer: an employer calendar is needed. An LLM should not guess the person's working schedule from the dates.

Privacy exercise

Create analyst and employer roles in the application. The analyst can read synthetic review guidance; the employer view returns only certificate status and permitted dates. Enforce the scope before retrieval and again before returning source excerpts. Redacting the final answer alone is insufficient because sensitive chunks may already have reached the model.

Chunking exercise

Compare a chunk containing only the sentence about overlapping leave with one containing the amendment exception and duration definition. Test whether the model distinguishes certificate-days from distinct absence days. A larger chunk can improve completeness but may add unrelated context and increase token cost.

Interview Q1 | How do you handle dates?

Normalize dates and preserve time zones for timestamps. Define inclusive versus exclusive intervals explicitly, and use a deterministic function for interval operations. Store both event time and ingestion time where historical reconstruction matters.

Interview Q2 | What should the system do when policy is absent?

Say that the supplied knowledge base does not establish the answer, identify what policy is missing and avoid fabricating a rule. This is a successful abstention, not necessarily a retrieval failure. Interview Q3: How do you test privacy? Use role-specific queries, direct source access attempts, adversarial prompts and logs that avoid storing unnecessary sensitive text.

Advanced extension

Add version-aware retrieval and authorized structured lookups for certificate status. The retriever supplies policy; a validated query computes counts from records. Keep tool outputs and policy citations distinct. See FWA-06 for tool safety, evaluation slices and release criteria.